Postulates, Theorems, Converses and Biconditionals Answer Key

High School Geometry   Definitions, Postulates, and Proof Logic

Quick Answers

1. 5	4. -6, 6	7. —	10. —
2. —	5. 8	8. 6	
3. 13	6. 15	9. —	

1. Segment Addition.

(a) The **Segment Addition Postulate**: if B is between A and C, then $AB+BC=AC$. It is a *postulate* — betweenness on a line is assumed, not derived, so nothing earlier is available to prove it.

(b) Substitute and collect:

$$(2x+3)+(3x-1)=27 \implies 5x+2=27 \implies 5x=25$$

$$\boxed{x=5}$$

Check: $AB=13$, $BC=14$, and $13+14=27$.

2. Classifying five statements.

Step 1. The deciding question for each row is "could this be proved from something earlier, and is it reversible?"

Step 2. Row by row:

- *Through any two points there is exactly one line* — **postulate**; it is a starting assumption about lines and points.
- *An angle bisector divides an angle into two congruent angles* — **definition**; it states what the term means, and it is reversible.
- *Vertical angles are congruent* — **theorem**; it is proved from the Linear Pair Postulate and angle subtraction.
- *If M is the midpoint of $\overline{AB}$, then $AM=MB$* — **definition** (one direction of the definition of midpoint); nothing is proved, the term is being unpacked.
- *The angles of a triangle sum to* $180°$ — **theorem**; its proof uses the Parallel Postulate.

Grading note: accept "definition" or "postulate" on the midpoint row only when the reason given is that no proof is involved. The reason column carries the learning here, not the label. *Manual review: read the student's reasons.*

3. Angle Addition.

(a) The **Angle Addition Postulate**: if D lies in the interior of $\angle ABC$, then $m\angle ABD + m\angle DBC = m\angle ABC$. Like segment addition, it is assumed rather than proved.
(b) Substitute and collect:

$$4x + (3x + 5) = 96 \implies 7x + 5 = 96 \implies 7x = 91$$

$$\boxed{x = 13}$$

Check: $m\angle ABD = 52°$ and $m\angle DBC = 44°$, and $52 + 44 = 96$.

4. A converse that fails.

(a) Converse: *if* $x^2 = 36$*, then* $x = 6$. (Hypothesis and conclusion swap places; nothing else changes.)
(b) Solve completely — "completely" is the whole point of the problem:

$$x^2 - 36 = 0 \implies (x - 6)(x + 6) = 0$$

$$\boxed{x = -6 \text{ or } x = 6}$$

(c) The converse is **false**. Taking $x = -6$ gives $x^2 = 36$ (hypothesis true) but $x \neq 6$ (conclusion false), which is exactly what a counterexample is. The original statement stays true — a conditional and its converse are independent claims.

5. Testing a biconditional.

(a) The two conditionals are: *if* $2x - 7 = 9$ *then* $x = 8$, and *if* $x = 8$ *then* $2x - 7 = 9$.
(b) Solve: $2x - 7 = 9 \Rightarrow 2x = 16$. $\boxed{x = 8}$
The solution set is exactly $\{8\}$.
(c) The forward direction holds because 8 is the *only* solution — if there were a second one, the equation could be satisfied by something other than 8 and the forward conditional would fail. The backward direction holds because substituting 8 gives $16 - 7 = 9$. Both directions true means the biconditional is valid. *This is the general test: a biconditional is valid exactly when the solution sets on the two sides are the same set.*

6. The converse of the Vertical Angles Theorem.

(a) Converse: *if two angles are congruent, then they are vertical angles.*
(b) Congruent means equal measure:

$$5x - 8 = 3x + 22 \implies 2x = 30$$

$$\boxed{x = 15}$$

Then $m\angle 1 = 5(15) - 8 = 67°$ and $m\angle 2 = 3(15) + 22 = 67°$.
(c) These two angles are congruent (both 67°) but they are adjacent, not vertical. So the hypothesis of the converse holds while its conclusion fails — one counterexample, and the converse is **false**. The theorem itself remains true; only the reversed statement is being tested.

7. Why one is a postulate and the other a theorem.

Step 1. Obviousness is not the criterion — *provability from what came before* is. A postulate is a place the system starts; a theorem is somewhere it arrives.
Step 2. "Through any two points there is exactly one line" has nothing earlier to appeal to: it is one of the assumptions that gives the words *point* and *line* their behaviour.
Step 3. "Two distinct lines intersect in exactly one point" can be argued: suppose two distinct lines met at two different points. Those two points would then lie on two different lines, contradicting the postulate that two points determine exactly one line. So the theorem rests on the postulate above, and its proof is an indirect one.
Manual review: accept any explanation that (i) says postulates are assumed and theorems are proved, and (ii) names the two-points-determine-a-line postulate as what the proof leans on.

8. Perpendicular bisector, both directions.

(a) The biconditional says P is on the perpendicular bisector exactly when $PA = PB$, so set the two distances equal:

$$4x - 5 = 2x + 7 \implies 2x = 12$$

$$\boxed{x = 6}$$

(b) $PA = 4(6) - 5 = 19$ and $PB = 2(6) + 7 = 19$. They agree, as they must.
(c) The half used is *if $PA = PB$, then P is on the perpendicular bisector* — the work started from equal distances and concluded a position. The other half runs the other way: given that P is on the bisector, conclude $PA = PB$. Because the statement is a biconditional, both halves are available, which is what makes it worth stating as one sentence instead of two.

9. Converse of the square statement.

(a) Converse: *if a quadrilateral has four right angles, then it is a square.*
(b) **False.**
(c) Counterexample: the rectangle with vertices $(0, 0)$, $(6, 0)$, $(6, 3)$, $(0, 3)$. All four of its angles are right angles, so the hypothesis holds. But its sides measure 6, 3, 6, 3, so it fails the part of the definition of a square that requires *all four sides congruent.* Any non-square rectangle works; the labelled side lengths are what make it an argument rather than a picture.
(d) Because the converse is false, "a quadrilateral is a square if and only if it has four right angles" is **not** a valid biconditional. Only the forward direction survives. A correct biconditional needs a condition that pins down the shape completely, e.g. *a quadrilateral is a square if and only if it has four right angles and four congruent sides.*
Manual review: read the student's figure and reason; accept any labelled non-square rectangle.

10. Definitions reverse; other true statements may not.

(a) As a biconditional: *a ray is an angle bisector of an angle if and only if it divides that angle into two congruent angles.* Both directions are true because a definition is an agreement about the meaning of the term — that reversibility is what makes definitions the only statements you may use "both ways" without proof.

(b) No. Test each direction separately:

Forward — if two angles are complementary, then they are adjacent: **false**. A 30° angle drawn on this page and a 60° angle drawn on another page sum to 90° and share nothing at all.

Converse — if two angles are adjacent, then they are complementary: **false**. Two adjacent angles measuring 100° and 40° share a side and a vertex but sum to 140°.

Both directions fail, so no biconditional can be formed. Complementary is about *measure* and adjacent is about *position*, and the two ideas are independent — which is precisely why the definition of complementary says nothing about position.

Manual review: accept any pair of counterexamples that separates measure from position.